MultiPoseNet: Multi-Person Pose Estimation using Pose Residual Network

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Introduction

In this paper, we present a new bottom-up method for multi-person 2D pose estimation. MultiPoseNet can jointly handle person detection, person segmentation and pose estimation problems. The novel assignment method is implemented by the *Pose Residual Network (PRN)* which receives keypoint and person detections, and produces accurate poses by assigning keypoints to person instances. On the COCO keypoints dataset, our pose estimation method outperforms all previous bottom-up methods both in accuracy (+4-point mAP over previous best result) and speed. Our main contributions are:

- We propose the Pose Residual Network (PRN), a simple yet very effective method for the problem of assigning/grouping body joints.
- We outperform all previous bottom-up methods and achieve comparable performance with top-down methods.
- Our network architecture is extendible; we show that using the same backbone, one can solve other related problems, too, e.g. person segmentation.
- Our method works faster than all previous methods, in real-time at 23 frames/sec.

(a) (b) (c) (d)

Figure 2: Bounding box overlapping scenarios.

Methods

- **Shared Backbone** serves as a feature extractor for keypoint and person detection subnets. It is actually a ResNet with two Feature Pyramid Networks (one for the keypoint subnet, the other for the person detection subnet) connected to it

- **Keypoint Subnet** outputs keypoint and segmentation heatmaps. Heatmaps represent keypoint locations as Gaussian peaks. Each heatmap layer belongs to a specific keypoint class (nose, wrists, ankles etc.) and contains arbitrary number of peaks that pertain to person instances.

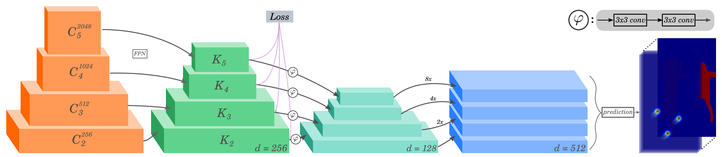

Figure 3: The architecture of the keypoint subnet. It takes hierarchical CNN features as input and outputs keypoint and segmentation heatmaps.

- **Person Detection Subnet** produces person bounding boxes, using single stage RetinaNet model.

- **Pose Residual Network:** Assigning keypoint detections to person instances (bounding boxes, in our case) is straightforward if there is only one person in the bounding box. However, it becomes non-trivial if there are overlapping people in a single box. In the case of an overlap, a bounding box can contain multiple keypoints not related to the person in question, and this creates ambiguity in constructing final pose predictions. We solve these ambiguities using Pose Residual Network.

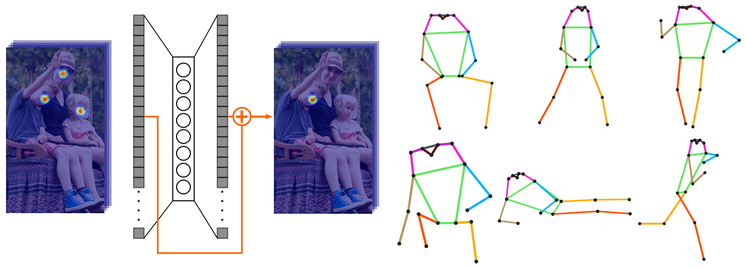

Figure 4: Left: Pose Residual Network (PRN). The PRN is able to disambiguate which keypoint should be assigned to the current person box. **Right:** Six sample poses obtained via clustering the structures learned by PRN.

Let $\mathbf{X}$ denote the input to the PRN, where $\mathbf{X} = \{\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_k\}$ in which $\mathbf{x}_k \in \mathbb{R}^{W \times H}$, k is the number of different keypoint types. The final goal of PRN is to output $\mathbf{Y}$ where $\mathbf{Y} = \{\mathbf{y}_1, \mathbf{y}_2, \ldots, \mathbf{y}_k\}$, in which $\mathbf{y}_k \in \mathbb{R}^{W \times H}$ is of the same size as $\mathbf{x}_k$, containing the correct position for each keypoint indicated by a peak in that keypoints channel. PRN models the mapping from $\mathbf{X}$ to $\mathbf{Y}$ as

$$\mathbf{y}_k = \phi_k(\mathbf{X}) + \mathbf{x}_k \tag{1}$$

where the functions $\phi_1(\cdot), \ldots, \phi_K(\cdot)$ apply a *residual correction* to the pose in $\mathbf{X}$, hence the name pose residual network.

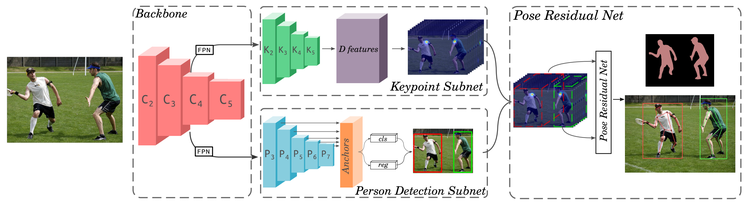

Figure 5: MultiPoseNet is a multi-task learning architecture capable of performing human keypoint estimation, detection and semantic segmentation tasks altogether efficiently.

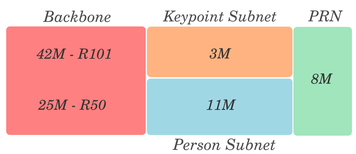

Figure 6: Number of parameters for each block of MultiPoseNet.

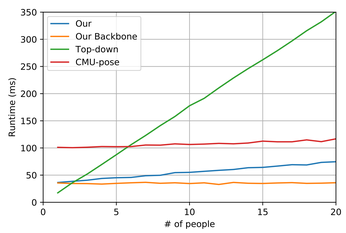

Figure 7: Runtime analysis of MultiPoseNet with respect to number of people.

Results

Table 1: Results on COCO **test-dev**, excluding systems trained with external data. Top-down methods are shown separately to make a clear comparison between bottom-up methods.

		FPS	AP	AP_{50}	AP_{75}	AP_M	AP_L	AR	AR_{50}	AR_{75}	AR_M	AR_L
BU	Ours	23	**69.6**	86.3	**76.6**	**65.0**	**76.3**	**73.5**	88.1	**79.5**	**68.6**	**80.3**
BU	Newell et al.	6	65.5	**86.8**	72.3	60.6	72.6	70.2	**89.5**	76.0	64.6	78.1
BU	CMU-Pose	10	61.8	84.9	67.5	57.1	68.2	66.5	87.2	71.8	60.6	74.6
TD	Megvii	-	73.0	91.7	80.9	69.5	78.1	79.0	95.1	85.9	74.8	84.6
TD	CFN	3	72.6	86.7	69.7	78.3	64.1	-	-	-	-	-
TD	Mask R-CNN	5	69.2	90.4	76.0	64.9	76.3	75.2	93.7	81.1	70.3	81.8
TD	SJTU	0.4	68.8	87.5	75.9	64.6	75.1	73.6	91.0	79.8	68.9	80.2
TD	GRMI-2017	-	66.9	86.4	73.6	64.0	72.0	71.6	89.2	77.6	66.1	79.1
TD	G-RMI-2016	-	60.5	82.2	66.2	57.6	66.6	66.2	86.6	71.4	61.9	72.2

Table 2: Left: PRN assignment results with non-grouped keypoints obtained from two bottom-up methods. **Right:** Ablation experiments of PRN with COCO validation data.

Models	AP	AP_{50}	AP_{75}	AP_M	AP_L
PAF	58.4	81.5	62.6	**54.4**	65.1
PRN + PAF	**59.2**	**82.2**	**64.4**	54.1	**67.0**
SHG	56.9	80.8	61.3	49.9	**68.8**
PRN + SHG	**58.1**	**81.4**	**63.0**	**51.3**	68.1

PRN Ablations	AP	AP_{50}	AP_{75}	AP_M	AP_L
Both GT	89.4	97.1	91.2	87.9	91.8
GT keypoints + Our bbox	75.3	82.1	78	70.1	84.5
Our keypoints + GT bbox	65.1	89.2	76.2	60.3	74.7
PRN	64.3	88.2	75	59.6	73.9
UCR	49.7	59.5	52.4	44.1	51.6
Max	45.3	55.1	48.8	40.6	46.9

Figure 8: Some qualitative results for COCO test-dev dataset.

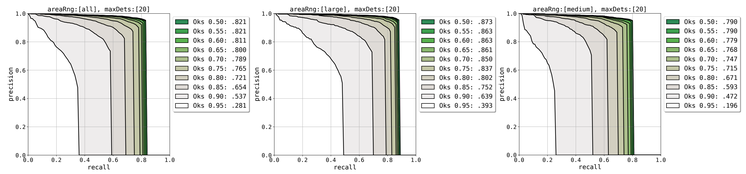

Figure 9: Precision-recall curves on COCO validation set across scales *all, large* and *medium*.

ArXiv paper: *https://arxiv.org/abs/1807.04067* **GitHub:** *https://github.com/mkocabas/pose-residual-network*